

Tribhuvan University
Faculty of Humanities & Social Sciences
Bachelor in Computer Applications (BCA) — Semester III
Course Title: Probability and Statistics (CAST 202)

COMPLETE QUESTION BANK WITH SOLUTIONS
Year-wise: 2019 | 2020 | 2021 | 2022 | 2023 | 2024

All questions solved with step-by-step working,
formulas, tables, and final answers.

Full Marks: 60 | Pass Marks: 24 | Time: 3 Hours

Group B [6×5 = 30] — Attempt any SIX questions

Q2. Describe scope and limitation of Statistics.

Answer:

Scope of Statistics:

Statistics has wide scope in almost every field of human knowledge and activity.

1. **Business & Economics:** Demand forecasting, quality control, market research.
2. **Natural Sciences:** Biological experiments, medical research, physics.
3. **Social Sciences:** Census, population studies, sociological surveys.
4. **Computer Science:** Data mining, machine learning, algorithm analysis.
5. **Government:** Planning, policy making, national accounting.
6. **Agriculture:** Crop yield analysis, soil testing experiments.

Limitations of Statistics:

1. Statistics deals only with **quantitative data** — qualitative facts (honesty, beauty) cannot be directly measured.
2. Statistics deals with **aggregates**, not individual items.
3. Statistical results are **true on average** — not necessarily for each specific case.
4. Statistics can be **misused** — wrong conclusions can be drawn if methods are misapplied.
5. It requires **trained experts** for correct interpretation.
6. It does not reveal the **complete picture** by itself.

Q3. Determine average wages from following data:

Wages	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70
No. of Worker	10	13	18	21	24	28	20	11	8

Answer (Arithmetic Mean by Step-Deviation Method):

$$\text{Formula: Mean } (\bar{X}) = A + (\Sigma fd / \Sigma f) \times h$$

Where: A = Assumed mean (mid-value of modal class), h = class width, d = (m - A)/h

Wages	f	Mid-val (m)	d=(m-47.5)/5	fd
25-30	10	27.5	-4	-40
30-35	13	32.5	-3	-39
35-40	18	37.5	-2	-36
40-45	21	42.5	-1	-21
45-50	24	47.5	0	0
50-55	28	52.5	1	28
55-60	20	57.5	2	40
60-65	11	62.5	3	33
65-70	8	67.5	4	32
Total	153			-3

$$\Sigma f = 153, \Sigma fd = 40+33+32+28 - 40-39-36-21 = 133 - 136 = -3$$

$$X_{\bar{}} = 47.5 + (-3/153) \times 5 = 47.5 - 0.098 \approx 47.40$$

∴ Average Wage ≈ Rs. 47.40

Q4. Calculate Karl Pearson's correlation coefficient from the following data:

Sales	43	41	36	34	50
Expenses	10	22	13	19	17

Answer:

$$\text{Formula: } r = [n \cdot \Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{[n \cdot \Sigma x^2 - (\Sigma x)^2][n \cdot \Sigma y^2 - (\Sigma y)^2]}$$

x	y	x ²	y ²	xy
43	10	1849	100	430
41	22	1681	484	902
36	13	1296	169	468
34	19	1156	361	646
50	17	2500	289	850
Σx=204	Σy=81	Σx ² =8482	Σy ² =1403	Σxy=3296

$$r = (5 \times 3296 - 204 \times 81) / \sqrt{[(5 \times 8482 - 204^2) \times (5 \times 1403 - 81^2)]}$$

$$r = -44 / \sqrt{[794 \times 454]}$$

$$r = -44 / 600.3965 \approx -0.0733$$

∴ r ≈ -0.073 — Negative correlation between Sales and Expenses.

Q5. Estimate the marks in JAVA when the marks in Statistics is 65 (Regression):

Marks in Statistics	57	58	59	59	60	61	62	64
Marks in JAVA	77	78	75	78	82	82	79	81

Answer (Regression of Y on X):

$$\text{Formula: } Y - \bar{Y} = b_{yx}(X - \bar{X}) \text{ where } b_{yx} = (n \cdot \Sigma xy - \Sigma x \cdot \Sigma y) / (n \cdot \Sigma x^2 - (\Sigma x)^2)$$

$$n=8, \Sigma x=480, \Sigma y=632, \Sigma x^2=28836, \Sigma xy=37944$$

$$\bar{X} = 60.000, \bar{Y} = 79.000$$

$$b_{yx} = (8 \times 37944 - 480 \times 632) / (8 \times 28836 - 480^2) = 192 / 288 = 0.6667$$

$$\text{Regression line: } Y - 79.000 = 0.6667(X - 60.000)$$

$$\text{When } X = 65: Y = 79.000 + 0.6667 \times (65 - 60.000) \approx 82.33$$

∴ Estimated Marks in JAVA = **82.33**

Q6. Fit Binomial Distribution (p = 0.5):

No. of heads	0	1	2	3	4
Frequency	28	62	46	20	4

Answer:

$$\text{Formula: } P(X=r) = C(n,r) \cdot p^r \cdot q^{(n-r)}, \text{ Expected freq} = N \times P(X=r)$$

$$\text{Here } p = q = 0.5, n = 4, N = 28+62+46+20+4 = 160$$

r	C(4,r)	P(X=r)	Expected Freq = 160×P
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0	1	0.0625	10.0
1	4	0.2500	40.0
2	6	0.3750	60.0
3	4	0.2500	40.0
4	1	0.0625	10.0

The expected frequencies closely match observed frequencies — Binomial fits well.

Q7. How do you determine sample size in sampling? Explain briefly.

Answer:

Formula for Sample Size (estimating mean):

$$n = (Z \cdot \sigma / E)^2$$

Where: Z = Z-value at desired confidence level, σ = population standard deviation, E = maximum allowable error (margin of error)

Formula for Sample Size (estimating proportion):

$$n = Z^2 \cdot p \cdot q / E^2 \text{ where } q = 1 - p$$

- At 90% confidence: $Z = 1.645$
- At 95% confidence: $Z = 1.96$
- At 99% confidence: $Z = 2.576$
- Larger $Z \rightarrow$ larger sample needed.
- Smaller E (less error allowed) $\rightarrow$ larger sample needed.
- Larger $\sigma \rightarrow$ larger sample needed.

Q8. Write short notes on simple random sampling.

Answer:

Definition: Simple Random Sampling (SRS) is a method of selecting a sample from a population such that every individual and every possible sample of size n has an equal and independent chance of being selected.

Types:

1. **SRS with Replacement (SRSWR):** Selected unit is returned before next draw; same unit can appear again.
2. **SRS without Replacement (SRSWOR):** Selected unit is not returned; each unit selected at most once.

Methods of Selection: Lottery method, Random number tables, Computer-generated random numbers.

Advantages: Free from bias, easy to apply, mathematically simple.

Disadvantages: Not suitable for very large populations; may not represent all sub-groups.

Group C [2×10 = 20] — Attempt any TWO questions

Q9. Student's age in Regular BCA and Morning BCA — which group is easier to teach? (CV comparison)

Regular BCA		Morning BCA	
Age	No. Students	Age	No. Students

23	9	27	10
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11
27	3	36	2
24	9	32	4
Total	60	Total	60

Answer (Coefficient of Variation comparison):

Formula: Mean $\bar{X} = \sum fx / \sum f$, Variance $\sigma^2 = \sum f(x - \bar{X})^2 / \sum f$, CV = $(\sigma / \bar{X}) \times 100\%$

The group with **LOWER CV is more homogeneous** → easier to teach.

Regular BCA: $\bar{X} = 24.45$, $\sigma = 2.24$, CV = 9.16%

Morning BCA: $\bar{X} = 31.20$, $\sigma = 2.97$, CV = 9.50%

∴ Regular BCA group has lower CV → more homogeneous in age → EASIER TO TEACH.

Q10. Normal Distribution: mean=200, s.d.=20. Find various probabilities:

Answer:

Formula: $Z = (X - \mu) / \sigma$, $\mu = 200$, $\sigma = 20$

Use Standard Normal Table (Z-table) where $\Phi(z) = P(Z \leq z)$

Part	Z-calculation	Working	Answer
i) $P(X > 180)$	$Z = (180 - 200) / 20 = -1.00$	$P(X > 180) = P(Z > -1) = P(Z < 1) = 0.8413$	0.8413
ii) $P(X < 220)$	$Z = (220 - 200) / 20 = 1.00$	$P(X < 220) = \Phi(1.00) = 0.8413$	0.8413
iii) $P(160 < X < 240)$	$Z_1 = (160 - 200) / 20 = -2$, $Z_2 = (240 - 200) / 20 = 2$	$P(-2 < Z < 2) = 2 \times 0.4772 = 0.9544$	0.9544
iv) $P(X > 220)$	$Z = (220 - 200) / 20 = 1.00$	$P(X > 220) = 1 - 0.8413 = 0.1587$	0.1587
v) $P(X < 180 \text{ or } X > 220)$	$P(X < 180) = 0.1587$, $P(X > 220) = 0.1587$	$P = 0.1587 + 0.1587 = 0.3174$	0.3174
vi) 10% values less than X	$P(Z < z) = 0.10 \rightarrow z = -1.282$	$X = 200 + (-1.282 \times 20) = 200 - 25.64 = 174.36$	$X = 174.36$

Q11. Labor Productivity Index — ANOVA (Two-Way):

Sector	2015	2016	2017
Agriculture	100	125	138
Manufacturing	100	60	53
Community & Social	100	89	80

Answer (Two-Way ANOVA):

H₀(Rows): No significant difference due to sector

H₀(Cols): No significant difference due to time period

Formulas:

$$CF = T^2/N, SSR = (\text{Row totals}^2/c) - CF, SSC = (\text{Col totals}^2/r) - CF$$

$$SST = \sum \sum x_{ij}^2 - CF, SSE = SST - SSR - SSC$$

Source	SS	df	MS	F-cal	F-tab(5%)
Rows (Sector)	3830.22	2	1915.11	3.71	5.14
Cols (Year)	169.56	2	84.78	0.16	5.14
Error	2063.11	4	515.78		
Total	6062.89	8			

$$CF = 845^2/9 = 79336.11$$

Row sums: [363, 213, 269], Col sums: [300, 274, 271]

Since F-row (3.71) < F-tab (5.14): Do not reject H0 → No significant difference by sector.

Since F-col (0.16) < F-tab (5.14): Do not reject H0 → No significant difference by year.

Group B [6×5 = 30] — Attempt any SIX questions

Q2. Write down the process of collecting primary data.

Answer:

Primary data is original data collected directly for a specific purpose. Methods:

1. **Direct Personal Interview:** Investigator meets respondent face-to-face. Accurate but costly.
2. **Indirect Oral Interview:** Information collected through witnesses/third parties.
3. **Mailed Questionnaire:** Printed questionnaire sent by post. Cheap, large coverage, but low response rate.
4. **Schedules through Enumerators:** Trained enumerators fill schedules. Reliable.
5. **Telephone Interview:** Quick, but limited to phone users.
6. **Online Survey:** Web-based forms — modern, fast, wide reach.

Q3. Determine Q1, D7, and P80 from the following data:

Age in year	10	12	14	16	18	20	22	24	26
No. of people	7	11	24	35	27	17	11	8	5

Answer:

Formulas:

$$Q1 = L + [(N/4 - cf)/f] \times h$$

$$D7 = L + [(7N/10 - cf)/f] \times h$$

$$P80 = L + [(80N/100 - cf)/f] \times h$$

Where: L=lower boundary, cf=cumulative freq before class, f=freq of class, h=class width

Age	f	CF
10	7	7
12	11	18
14	24	42
16	35	77
18	27	104
20	17	121
22	11	132
24	8	140
26	5	145

N = 145

Q1: $N/4 = 145/4 = 36.25 \rightarrow$ lies in class 16 (cf before = 42? Let's recalculate carefully)

cf: 10→7, 12→18, 14→42, 16→77 ... $N/4=36.25$ falls in class 14 (cf before=18, f=24)

$$Q1 = 14 + [(36.25-18)/24] \times 2 = 14 + (18.25/24) \times 2 = 14 + 1.52 = \mathbf{15.52}$$

D7: $7N/10 = 101.5 \rightarrow$ falls in class 20 (cf before=77+27=104? cf up to 18=104)... cf:16→77, 18→104, so 101.5 in class 18

$$D7 = 18 + [(101.5-77)/27] \times 2 = 18 + (24.5/27) \times 2 = 18 + 1.81 = \mathbf{19.81}$$

P80: $80N/100 = 116 \rightarrow$ cf: 20→121, cf before=104, f=17, class=20

$$P80 = 20 + [(116-104)/17] \times 2 = 20 + (12/17) \times 2 = 20 + 1.41 = \mathbf{21.41}$$

Q4. Correlation coefficient between income and expenditure (Kathmandu):

Income (000Rs)	10	11	12	13	14
Expenditure (000Rs)	9	8	9	12	11

Answer:

Formula: $r = [n \cdot \sum xy - \sum x \cdot \sum y] / \sqrt{[n \cdot \sum x^2 - (\sum x)^2][n \cdot \sum y^2 - (\sum y)^2]}$

x	y	x ²	y ²	xy
10	9	100	81	90
11	8	121	64	88
12	9	144	81	108
13	12	169	144	156
14	11	196	121	154
60	49	730	491	596

$$r = 40 / \sqrt{(50 \times 54)} = 40 / 51.9615 \approx \mathbf{0.7698}$$

∴ r ≈ 0.770 — Strong positive correlation between income and food expenditure.

Q5. Box contains 50 items, 20 defective. Probability of non-defective item selected:

Answer:

Formula: $P(\text{Event}) = \text{Favourable outcomes} / \text{Total outcomes}$

Total items = 50, Defective = 20, Non-defective = 50 – 20 = 30

$$P(\text{non-defective}) = 30/50 = 3/5 = \mathbf{0.60 \text{ or } 60\%}$$

Q6. What is sampling? Standard deviation = 0.5, 95% confidence, error ≤ 0.01. Find sample size.

Answer:

Sampling: Sampling is the process of selecting a representative subset (sample) from a population to draw conclusions about the whole population.

Formula: $n = (Z \cdot \sigma / E)^2$

Z = 1.96 (at 95% confidence), σ = 0.5, E = 0.01

$$n = (1.96 \times 0.5 / 0.01)^2 = (98)^2 = \mathbf{9604}$$

∴ Required sample size = **9604**

Q7. Calculate the Median and Mode from the following distribution:

Expenditure (000Rs)	10-20	20-30	30-40	40-50	50-60	60-70
Number of families	7	12	15	13	8	5

Answer:

Median Formula: $Md = L + [(N/2 - cf)/f] \times h$

Mode Formula: $Mo = L + [f1 - f0 / (2f1 - f0 - f2)] \times h$

Class	f	CF
10-20	7	7

20-30	12	19
30-40	15	34
40-50	13	47
50-60	8	55
60-70	5	60

$N = 60, N/2 = 30.0$

$N/2 = 30 \rightarrow$ falls in class 30-40 (cf before = 19, $f = 15$)

Median = $30 + [(30-19)/15] \times 10 = 30 + 7.33 = \mathbf{37.33 (000Rs)}$

Modal class = 30-40 (highest frequency $f_1=15$), $f_0=12$, $f_2=13$

Mode = $30 + [(15-12)/(2 \times 15 - 12 - 13)] \times 10 = 30 + [3/5] \times 10 = 30 + 6 = \mathbf{36 (000Rs)}$

Q8. One-way ANOVA — test scores from 3 provinces (Gandaki, Lumbini, Bagmati):

Gandaki	9	7	6
Lumbini	7	4	5
Bagmati	6	5	6

Answer (One-Way ANOVA):

H0: $\mu_1=\mu_2=\mu_3$ (no significant difference among provinces)

H1: At least one mean differs

Formulas: $CF = T^2/N$, $SSB = \sum(T_i^2/n_i) - CF$, $SSW = SST - SSB$

$MSB = SSB/(k-1)$, $MSW = SSW/(N-k)$, $F = MSB/MSW$

$T = 55, N = 9, k = 3$

$CF = 55^2/9 = 336.11$

$T_{\text{Gandaki}}=22, T_{\text{Lumbini}}=16, T_{\text{Bagmati}}=17$

$SSB = 22^2/3 + 16^2/3 + 17^2/3 - 336.11 = 6.89$

$SST = 16.89, SSW = 10.00$

Source	SS	df	MS	F-cal	F-tab(5%, 2,6)
Between	6.89	2	3.44	2.07	5.14
Within	10.00	6	1.67		
Total	16.89	8			

F-cal (2.07) < F-tab (5.14) $\rightarrow$ Fail to reject H0 $\rightarrow$ No significant difference among provinces.

Group C [2×10 = 20] — Attempt any TWO questions

Q9. Average marks, SD and CV from grouped data:

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of people	54	90	86	58	62	82	78	66	70

Answer:

Formulas: $X_{\square} = A + (\sum fd / \sum f) \times h$, $\sigma^2 = [\sum fd^2 / \sum f - (\sum fd / \sum f)^2] \times h^2$, $CV = (\sigma / X_{\square}) \times 100$

Marks	f	Mid (m)	d=(m-45)/10	fd	fd ²
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0-10	54	5	-4	-216	864
10-20	90	15	-3	-270	810
20-30	86	25	-2	-172	344
30-40	58	35	-1	-58	58
40-50	62	45	0	0	0
50-60	82	55	1	82	82
60-70	78	65	2	156	312
70-80	66	75	3	198	594
80-90	70	85	4	280	1120
Total	646			0	4184

$$\bar{X} = 45 + (0/646) \times 10 = 45.00$$

$$\sigma^2 = (4184/646 - (0/646)^2) \times 100 = 647.68$$

$$\sigma = \sqrt{647.68} = 25.45$$

$$CV = (25.45/45.00) \times 100 = 56.55\%$$

Q10. Normal Distribution: mean=200, SD=20. Find probabilities: P(X>180), P(X<220), P(160<X<240), 10% values less than what X?

Answer: [Same as 2019 Q10 — see Set 1 Q10 above for detailed solution]

$\mu=200, \sigma=20$. Use $Z = (X-\mu)/\sigma$

Part	Z	Result
a) P(X>180)	Z=-1.00	P=0.8413
b) P(X<220)	Z=1.00	P=0.8413
c) P(160<X<240)	Z=±2.00	P=0.9544
d) P(X>220)	Z=1.00	P=0.1587
e) 10% < X	Z=-1.282	X=174.36

Q11. Describe simple random sampling with a suitable example.

Answer: [See 2019 Q8 for full details]

Example: Suppose we want to select 5 students from a class of 50. Assign numbers 01–50 to all students. Use a random number table or lottery to pick 5 numbers. The students corresponding to these numbers form the simple random sample.

Group B [6×5 = 30] — Attempt any SIX questions

Q11. Discuss the role of statistics in computer application.

Answer:

1. **Data Mining & Machine Learning:** Statistical methods underpin algorithms for finding patterns in large datasets.
2. **Algorithm Analysis:** Statistical techniques evaluate algorithm efficiency and performance.
3. **Software Testing:** Statistical testing (e.g., hypothesis tests) determines if software meets quality standards.
4. **Database Management:** Query optimization, indexing use statistical properties of data distributions.
5. **Image & Signal Processing:** Statistical filtering, noise removal, compression algorithms.
6. **Artificial Intelligence:** Bayesian networks, neural networks, probabilistic models.
7. **Network Traffic Analysis:** Statistical models predict network load and bottlenecks.
8. **Cryptography:** Random number generation and statistical tests for encryption strength.

Q12. Monthly salary of employees in Lalitpur Sub-metropolitan City — find mean:

Salary (000)	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No of employees	15	22	28	35	24	20	16

Answer (Mean by Step-Deviation Method):

$$\text{Formula: } X_{\bar{}} = A + (\Sigma fd / \Sigma f) \times h$$

Salary	f	Mid (m)	d=(m-35)/10	fd
0-10	15	5	-3	-45
10-20	22	15	-2	-44
20-30	28	25	-1	-28
30-40	35	35	0	0
40-50	24	45	1	24
50-60	20	55	2	40
60-70	16	65	3	48
Total	160			-5

$$X_{\bar{}} = 35 + (-5/160) \times 10 = 35 + -0.31 = 34.69 \text{ (000 Rs)}$$

Q13. Karl Pearson's correlation between Advertising Cost and Sales:

Advertising Cost (000)	39	65	62	90	82	75	25	98	36
Sales costs (000)	47	53	58	86	62	68	60	91	95

Answer:

$$\text{Formula: } r = [n \cdot \Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{\{[n \cdot \Sigma x^2 - (\Sigma x)^2][n \cdot \Sigma y^2 - (\Sigma y)^2]\}}$$

x	y	x ²	y ²	xy
39	47	1521	2209	1833
65	53	4225	2809	3445

62	58	3844	3364	3596
90	86	8100	7396	7740
82	62	6724	3844	5084
75	68	5625	4624	5100
25	60	625	3600	1500
98	91	9604	8281	8918
36	95	1296	9025	3420
572	620	41564	45152	40636

$$r = 11084 / 32095.54 \approx 0.3453$$

$\therefore r \approx 0.345$ — Positive correlation between advertising cost and sales.

Q14. Regression of maintenance cost on age of computer. Estimate cost for 10 yrs:

Age of computers (yrs)	2	4	6	8	10
Maintenance cost (00)	10	15	22	32	46

Answer (Regression Y on X):

$$\text{Formulas: } b = (n \cdot \Sigma xy - \Sigma x \cdot \Sigma y) / (n \cdot \Sigma x^2 - (\Sigma x)^2), a = \bar{y} - b \cdot \bar{x}$$

x	y	x ²	xy
2	10	4	20
4	15	16	60
6	22	36	132
8	32	64	256
10	46	100	460
30	125	220	928

$$b = (5 \times 928 - 30 \times 125) / (5 \times 220 - 30^2) = 890/200 = 4.4500$$

$$a = 25.00 - 4.4500 \times 6.00 = -1.7000$$

Regression Line: $Y = -1.7000 + 4.4500X$

$$\text{When } X = 10: Y = -1.7000 + 4.4500 \times 10 = 42.80 \text{ (00 Rs)}$$

Q15. Fit Poisson Distribution and find expected frequencies:

X	0	1	2	3	4	5	6	7	8
f	56	156	132	92	37	22	4	0	1

Answer:

$$\text{Formula: } P(X=r) = e^{(-\lambda)} \cdot \lambda^r / r!, \text{ where } \lambda = \Sigma fx / \Sigma f$$

$$N = 500, \lambda = \Sigma fx / \Sigma f = 986/500 = 1.9720$$

$$e^{(-1.9720)} \approx 0.13918$$

r	P(X=r)	Expected Freq = N×P
0	0.13918	69.59
1	0.27446	137.23
2	0.27062	135.31

3	0.17789	88.94
4	0.08770	43.85
5	0.03459	17.29
6	0.01137	5.68
7	0.00320	1.60
8	0.00079	0.39

Q16. Determine Q1, D7, P80 from age data:

Age in years	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No of peoples	10	11	27	35	29	17	11	10

Answer:

Formulas: $Q1 = L + [(N/4 - cf)/f] \times h$, $D7 = L + [(7N/10 - cf)/f] \times h$, $P80 = L + [(80N/100 - cf)/f] \times h$

Class	f	CF
10-20	10	10
20-30	11	21
30-40	27	48
40-50	35	83
50-60	29	112
60-70	17	129
70-80	11	140
80-90	10	150

$N = 150$

Q1: $N/4 = 37.5$. cf: 10-20→10, 20-30→21, 30-40→48. $N/4=37.5$ falls in 30-40 (cf before=21, f=27)

$$Q1 = 30 + (37.5 - 21)/27 \times 10 = \mathbf{36.11}$$

D7: $7N/10 = 105.0$. Falls in 40-50 (cf before=48, f=35)

$$D7 = 40 + (105.0 - 48)/35 \times 10 = \mathbf{56.29}$$

P80: $80N/100 = 120.0$. Falls in 60-70 (cf before=111, f=17)

$$P80 = 60 + (120.0 - 111)/17 \times 10 = \mathbf{65.29}$$

Q17. Box has 30 items, 10 defective. Two items selected WITHOUT replacement. Find P(both defective) and P(both non-defective).

Answer:

Formula: $P(A \cap B) = P(A) \times P(B|A)$ [for without replacement]

Total = 30, Defective = 10, Non-defective = 20

(a) P(both defective):

$$P = (10/30) \times (9/29) = 90/870 = 3/29 \approx \mathbf{0.1034}$$

(b) P(both non-defective):

$$P = (20/30) \times (19/29) = 380/870 = 38/87 \approx \mathbf{0.4368}$$

Group C [2×10 = 20] — Attempt any TWO questions

Q18. Intelligence test scores — Section A vs Section B — which is more consistent? (CV test)

Student no.	1	2	3	4	5	6	7	8	9	10
Section A	9	8	10	6	7	6	7	8	9	10
Section B	10	8	6	8	9	8	7	8	5	8

Answer (CV comparison):

Formulas: $X = \sum x/n$, $\sigma = \sqrt{[\sum (x - X)^2/n]}$, $CV = (\sigma/X) \times 100$

Section A: $X = 8.00$, $\sigma = 1.4142$, $CV = 17.6777\%$

Section B: $X = 7.70$, $\sigma = 1.3454$, $CV = 17.4722\%$

$CV(B) = 17.47\% < CV(A) = 17.68\% \rightarrow$ **Section B is more consistent.**

Q19. Normal Distribution: mean=5000 hrs, SD=100 hrs. Find probabilities:

Answer: $\mu=5000$, $\sigma=100$, $Z=(X-5000)/100$

Part	Z-value	Result
a) $P(X < 3012)$	$Z = (3012 - 5000)/100 = -19.88$	≈ 0 (essentially zero)
b) $P(4000 < X < 6000)$	$Z_1 = -10, Z_2 = 10$	≈ 1.0000
c) $P(X < 4500)$	$Z = (4500 - 5000)/100 = -5.00$	≈ 0 (essentially zero)
d) $P(X > 7000)$	$Z = (7000 - 5000)/100 = 20.00$	≈ 0 (essentially zero)

Note: With $\sigma=100$, values at 3012, 4000, 4500, 7000 are many standard deviations away $\rightarrow$ probabilities are essentially 0 or 1.

Q20. Two-Way ANOVA (Varieties $\times$ Blocks):

Varieties	Block 1	Block 2	Block 3
A	10	9	8
B	7	7	7
C	8	5	4
D	5	4	4

Answer (Two-Way ANOVA):

H0(rows): No diff among varieties

H0(cols): No diff among blocks

$T=78$, $N=12$, $CF=507.00$

Row sums=[27, 21, 17, 13], Col sums=[30, 25, 23]

Source	SS	df	MS	F-cal
Rows (Variety)	35.67	3	11.89	14.76
Cols (Block)	6.50	2	3.25	4.03
Error	4.83	6	0.81	
Total	47.00	11		

$F_{tab}(3,6)$ at 5% = 4.76, $F_{tab}(2,6)$ at 5% = 5.14

F-rows=14.76 vs 4.76: Significant diff among varieties.

F-cols=4.03 vs 5.14: No significant diff.

Group B [6×5 = 30] — Attempt any SIX questions

Q11. Differentiate between Convenient Sampling and Systematic Random Sampling with examples.

Answer:

Aspect	Convenient Sampling	Systematic Random Sampling
Definition	Select units that are easiest to reach	Select every k-th unit from ordered list
Randomness	Non-random / biased	Partially random
Procedure	No fixed procedure	1. Order population 2. Select random start 3. Pick every k-th unit
Example	Interview first 50 people you meet	From 1000 students, pick every 10th (k=10)
Representativeness	Poor	Better than convenient
Cost	Very low	Low to moderate
Bias	High	Low (if no periodicity)

Q12. Find Mean, Median, then use empirical relation to find Mode:

Age group	10-20	20-30	30-40	40-50	50-60
No. of months (f)	10	35	17	12	9

Answer:

Empirical Relation: Mode = 3·Median – 2·Mean

N=83, N/2=41.5

N/2=41.5 falls in class 20-30 (cf before=10, f=35)

Class	f	Mid	CF	fm
10-20	10	15	10	150
20-30	35	25	45	875
30-40	17	35	62	595
40-50	12	45	74	540
50-60	9	55	83	495
Total	83			2655

Mean = 2655/83 = **31.99**

Median = $20 + \frac{(41.5-10)}{35} \times 10 = \mathbf{29.00}$

Mode = $3 \times 29.00 - 2 \times 31.99 = \mathbf{23.02}$

Q13. Mean=34.69, n=65. Find missing frequency:

Expense class (000 Rs)	10-20	20-30	30-40	40-50	50-60
No of hotels	9	12	25	10	?

Answer:

Let missing frequency = f5

Total N = 9+12+25+10+f5 = 65 → f5 = 65–56 = **9**

Verify: Mean = $(9 \times 15 + 12 \times 25 + 25 \times 35 + 10 \times 45 + 9 \times 55) / 65 = (135 + 300 + 875 + 450 + 495) / 65 = 2255 / 65 = 34.69 \checkmark$

Q14. Karl Pearson's correlation — Statistics vs Mathematics marks:

Students	A	B	C	D	E	F	G	H	I	J
Statistics	42	51	65	57	56	28	19	24	33	80
Mathematics	42	61	66	54	57	28	29	21	35	87

Answer:

Formula: $r = [n \cdot \Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{[n \cdot \Sigma x^2 - (\Sigma x)^2][n \cdot \Sigma y^2 - (\Sigma y)^2]}$

$n=10, \Sigma x=455, \Sigma y=480, \Sigma x^2=24185, \Sigma y^2=26866, \Sigma xy=25389$

$r = 35490 / \sqrt{(34825 \times 38260)} = 35490 / 36502.12 \approx 0.9723$

$\therefore r \approx 0.972$ — Very strong positive correlation.

Q15. 2% defective articles in 300/day. Poisson distribution: i) exactly 2 defective/day ii) at least 2/day

Answer:

Formula: $P(X=r) = e^{-\lambda} \cdot \lambda^r / r!$ where $\lambda = n \cdot p = 300 \times 0.02 = 6$

$\lambda = 300 \times 0.02 = 6.0, e^{-6} \approx 0.002479$

i) $P(X=2) = e^{-6} \cdot 6^2 / 2! = 0.002479 \times 36 / 2 = 0.044618 \approx 0.0446$

ii) $P(X \geq 2) = 1 - P(X=0) - P(X=1)$

$P(X=0) = 0.002479, P(X=1) = 0.014873$

$P(X \geq 2) = 1 - 0.002479 - 0.014873 = 0.982649$

Q16. Fit Binomial Distribution — 6 coins tossed 500 times, $p=0.21$:

X	0	1	2	3	4	5	6	Total
f	165	80	70	55	45	50	35	500

Answer:

Formula: $P(X=r) = C(6,r) \cdot (0.21)^r \cdot (0.79)^{6-r}$, Expected = $500 \times P(X=r)$

r	C(6,r)	$(0.21)^r$	$(0.79)^{6-r}$	P(X=r)	Expected
0	1	1.000000	0.243087	0.24309	121.5
1	6	0.210000	0.307706	0.38771	193.9
2	15	0.044100	0.389501	0.25765	128.8
3	20	0.009261	0.493039	0.09132	45.7
4	15	0.001945	0.624100	0.01821	9.1
5	6	0.000408	0.790000	0.00194	1.0
6	1	0.000086	1.000000	0.00009	0.0

Q17. Hospital: height normal, mean=53 inches, SD=3. Find $P(X < 57)$, $P(X > 64)$, $P(50 < X < 56)$:

Answer: $\mu=53, \sigma=3, Z=(X-53)/3$

Part	Z-calc	Working	Answer
i) $P(X < 57)$	$Z=(57-53)/3=1.33$	$\Phi(1.33)=0.9082$	0.9082

ii) $P(X > 64)$	$Z = (64 - 53)/3 = 3.67$	$1 - \Phi(3.67) = 1 - 0.9999 = 0.0001$	≈ 0.0001
iii) $P(50 < X < 56)$	$Z_1 = (50 - 53)/3 = -1, Z_2 = (56 - 53)/3 = 1$	$\Phi(1) - \Phi(-1) = 0.8413 - 0.1587$	0.6826

Group C [2×10 = 20] — Attempt any TWO questions

Q18. Linear regression — Automobile demand vs price. Find demand when price=25 lakhs:

Price of cars	12	13	40	15	16	70	80	90	15	77
Demands	24	28	12	13	10	6	5	4	11	5

Answer (Regression Y on X — demand on price):

Formulas: $b = (n \cdot \sum xy - \sum x \cdot \sum y) / (n \cdot \sum x^2 - (\sum x)^2)$, $a = \bar{y} - b \cdot \bar{x}$

x	y	x ²	y ²	xy
12	24	144	576	288
13	28	169	784	364
40	12	1600	144	480
15	13	225	169	195
16	10	256	100	160
70	6	4900	36	420
80	5	6400	25	400
90	4	8100	16	360
15	11	225	121	165
77	5	5929	25	385
428	118	27948	1996	3217

$$b = -18334/96296 = -0.19039$$

$$a = 11.8000 - (-0.19039) \times 42.8000 = 19.9488$$

$$\text{Regression: } Y = 19.9488 + (-0.19039)X$$

$$\text{When } X = 25: Y = 19.9488 + (-0.19039) \times 25 = 15.19$$

$R^2 = 0.5783 \rightarrow$ model explains 57.83% of variation in demand.

Q19. Two firms A and B — compare wages, find combined average:

	Firm A	Firm B
No of workers	200	300
CV of Daily Wages	5	6
Variance of wages	67	86

Answer:

Given: $n_1=200, n_2=300, CV_1=5, CV_2=6, \sigma_1^2=67, \sigma_2^2=86$

Step 1: Find SD and Mean

$$\sigma_1 = \sqrt{67} \approx 8.185, \sigma_2 = \sqrt{86} \approx 9.274$$

$$CV = (\sigma/X) \times 100 \rightarrow X = \sigma \times 100 / CV$$

$$X_1 = 8.1854 \times 100 / 5 = 163.7071$$

$$X_{12} = 9.2736 \times 100/6 = 154.5603$$

a) Wage Bill = $n \times X_{12}$

Firm A wage bill = $200 \times 163.71 = 32741.41$

Firm B wage bill = $300 \times 154.56 = 46368.09$

→ **Firm B has larger wage bill.**

b) Consistency: Lower CV → more consistent.

$CV(A)=5 < CV(B)=6$ → **Firm A has greater consistency.**

c) Combined Mean: $X_{12} = (n_1 \cdot X_{11} + n_2 \cdot X_{12}) / (n_1 + n_2)$

$$X_{12} = (200 \times 163.7071 + 300 \times 154.5603) / 500 = 158.2190$$

Q20. One-Way ANOVA — average sales of computers from 3 shops:

Shop A	10	15	18	21	35
Shop B	23	45	45	55	57
Shop C	45	67	78	78	90

Answer (One-Way ANOVA at 5% significance):

H₀: $\mu_A = \mu_B = \mu_C$ (no difference in average sales)

T=682, N=15, CF=31008.27

TA=99, TB=225, TC=358

Source	SS	df	MS	F-cal	F-tab(2,12)
Between	6709.73	2	3354.87	18.04	3.89
Within	2232.00	12	186.00		
Total	8941.73	14			

F-cal (18.04) > F-tab (3.89) → Reject H₀ → Significant difference in average sales.

Shop means: A=19.8, B=45.0, C=71.6 → **Shop C has greatest average sale.**

SET 5 — YEAR 2023

Group B [6×5 = 30] — Attempt any SIX questions

Q11. Discuss the role of statistics in computer application.

Answer: [Same as 2021 Q11 — see Set 3]

Key roles: Data Mining, ML, Algorithm Analysis, Software Testing, AI/Bayesian networks, Image Processing, Network Traffic Analysis, Cryptography.

Q12. Define Statistics. Telephone calls (200 intervals) — compute mean, median, mode:

No of Calls	0	1	2	3	4	5	6
Frequency	15	22	28	35	42	34	24

Answer:

Statistics: Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting numerical data to make decisions.

Formulas: Mean = $\Sigma fx / \Sigma f$, Median: find $N/2$ class, Mode: class with highest freq

x	f	CF	fx
0	15	15	0
1	22	37	22
2	28	65	56
3	35	100	105
4	42	142	168
5	34	176	170
6	24	200	144
Total	200		665

Mean = $665/200 = 3.3250$

$N/2 = 100.0$; cumulative freq reaches 100.0 at $x=3 \rightarrow$ **Median** = 3

Highest frequency = 42 at $x=4 \rightarrow$ **Mode** = 4

Q13. Define correlation. Calculate r between study hours and sleep hours. Find r^2 and interpret:

No of Study hrs	2	4	6	8	10
No of Sleep hrs	10	9	8	7	6

Answer:

Correlation: Correlation is a statistical measure that describes the strength and direction of the linear relationship between two variables.

Formula: $r = [n \cdot \Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{\{[n \cdot \Sigma x^2 - (\Sigma x)^2][n \cdot \Sigma y^2 - (\Sigma y)^2]\}}$

x	y	x^2	y^2	xy
2	10	4	100	20
4	9	16	81	36
6	8	36	64	48
8	7	64	49	56

10	6	100	36	60
30	40	220	330	220

$$r = -100 / \sqrt{(200 \times 50)} = -100/100.00 = -1.0000$$

$$r^2 \text{ (Coefficient of Determination)} = (-1.0000)^2 = 1.0000$$

Interpretation: $r=-1.00$ indicates perfect negative linear correlation. As study hours increase, sleep hours decrease perfectly. $r^2=1.00$ means 100% of variation in sleep hours is explained by study hours.

Q14. Define regression. Regression of cholesterol on age (n=10 men):

Age	58	69	43	39	63	52	47	31	74	36
Cholesterol	189	235	193	177	154	191	213	165	198	181

Answer:

Regression: Regression is a statistical method to find the best-fitting line (or curve) that describes the relationship between a dependent variable (Y) and one or more independent variables (X).

$$\text{Formula: } b = (n \cdot \Sigma xy - \Sigma x \cdot \Sigma y) / (n \cdot \Sigma x^2 - (\Sigma x)^2), a = \bar{y} - b \cdot \bar{x}$$

$$n=10, \Sigma x=512, \Sigma y=1896, \Sigma x^2=28110, \Sigma xy=98307$$

$$\bar{x}=51.20, \bar{y}=189.60$$

$$b = 12318/18956 = 0.6498$$

$$a = 189.60 - 0.6498 \times 51.20 = 156.3292$$

$$\text{Regression Line: Cholesterol} = 156.3292 + 0.6498 \times \text{Age}$$

Q15. Define Binomial Distribution. 70% chips get thick coating. n=15 chips. P(at least 12) and P(exactly 10):

Answer:

Binomial Distribution: A discrete probability distribution for n independent trials, each with probability p of success.

$$\text{Formula: } P(X=r) = C(n,r) \cdot p^r \cdot (1-p)^{(n-r)}, n=15, p=0.70, q=0.30$$

i) $P(X \geq 12) = P(X=12) + P(X=13) + P(X=14) + P(X=15)$:

r	C(15,r)	p^r	$q^{(15-r)}$	P(X=r)
12	455	0.013841	0.027000	0.170040
13	105	0.009689	0.090000	0.091560
14	15	0.006782	0.300000	0.030520
15	1	0.004748	1.000000	0.004748
P(X≥12)				0.296868

$$\therefore P(X \geq 12) = 0.296868$$

$$\text{ii) } P(X=10) = C(15,10) \cdot (0.7)^{10} \cdot (0.3)^5 = 3003 \times 0.028248 \times 0.002430 = 0.206130$$

Q16. Sample size: SD=0.95 sec, 95% confidence, error ≤ 0.01 sec:

Answer:

$$\text{Formula: } n = (Z \cdot \sigma / E)^2$$

$$Z=1.96 \text{ (95\%)}, \sigma=0.95, E=0.01$$

$$n = (1.96 \times 0.95 / 0.01)^2 = (186.20)^2 = 34670$$

$$\therefore \text{Required sample size} = 34671$$

Q17. Define sampling. Population = {1,3,4,8}. All samples of size 2 without replacement. Show sample mean is unbiased estimate.

Answer:

Sampling: Process of selecting a representative subset from a population.

Population: {1,3,4,8}, N=4, n=2, Samples = $C(4,2) = 6$

Sample	Sample Mean
(1, 3)	2.0
(1, 4)	2.5
(1, 8)	4.5
(3, 4)	3.5
(3, 8)	5.5
(4, 8)	6.0
Mean of sample means	4.0000

Population mean = $(1+3+4+8)/4 = 4.0000$

Mean of sample means = 4.0000 = Population mean = 4.0000

∴ Sample mean is an unbiased estimator of population mean. ✓

Group C [2×10 = 20] — Attempt any TWO questions

Q18. Two firms A and B — wage analysis, combined mean, combined variance:

	Firm A	Firm B
No. of workers	500	600
Average monthly wages	186	175
Variance	81	100

Answer:

i) Wage Bill = $n \times X$

Firm A: $500 \times 186 = 93000$

Firm B: $600 \times 175 = 105000$

→ **Firm B has larger wage bill.**

ii) CV: $\sigma_1 = \sqrt{81} = 9.0$, $\sigma_2 = \sqrt{100} = 10.0$

CV1 = $(9.0/186) \times 100 = 4.8387\%$, CV2 = $(10.0/175) \times 100 = 5.7143\%$

CV1 < CV2 → **Firm A has greater consistency (uniformity).**

iii) Combined Mean:

$$X_{12} = (500 \times 186 + 600 \times 175) / (500 + 600) = 198000 / 1100 = 180.0000$$

Combined Variance: $\sigma^2 = [n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)] / (n_1 + n_2)$

$$d_1 = X_{11} - X_{12} = 186 - 180.0000 = 6.0000$$

$$d_2 = X_{22} - X_{12} = 175 - 180.0000 = -5.0000$$

$$\sigma_{12}^2 = [500 \times (81 + 6.0000^2) + 600 \times (100 + (-5.0000)^2)] / 1100 = 121.3636$$

Q19. Pregnancy duration: normal, mean=266 days, SD=16. Find probabilities:

Answer: $\mu=266$, $\sigma=16$, $Z=(X-266)/16$

Part	Z-calculation	Working	Answer
i) $P(240 < X < 270)$	$Z_1 = (240 - 266)/16 = -1.625$, $Z_2 = (270 - 266)/16 = 0.25$	$1 - \Phi(-1.625) - \Phi(0.25) = 0.5987 - 0.5987 = 0$	0
ii) $P(X > 275)$	$Z = (275 - 266)/16 = 0.5625$	$1 - \Phi(0.5625) = 1 - 0.7131 = 0.2869$	0.2869
iii) $P(X < 272)$	$Z = (272 - 266)/16 = 0.375$	$\Phi(0.375) \approx 0.6462$	0.6462

Q20. Define ANOVA. One-way ANOVA test — 3 samples are equal?

Sample I	4	5	6	5
Sample II	7	5	6	5
Sample III	4	8	9	2

Answer:

ANOVA (Analysis of Variance): A statistical test used to determine whether there are statistically significant differences between the means of three or more independent groups.

$T = 66$, $N = 12$, $CF = 363.00$

$T_1 = 20$, $T_2 = 23$, $T_3 = 23$

Source	SS	df	MS	F-cal	F-tab(2,9) at 5%
Between	1.50	2	0.75	0.18	4.26
Within	37.50	9	4.17		
Total	39.00	11			

$F\text{-cal} (0.18) < F\text{-tab} (4.26) \rightarrow \text{Fail to reject } H_0 \rightarrow \text{No significant difference among sample means.}$

Group B [6×5 = 30] — Attempt any SIX questions

Q11. Describe the scope and limitations of Statistics. [5]**Answer:** [See 2019 Q2 for detailed answer — same question]

Scope covers: Business, Economics, Natural Sciences, Social Sciences, Computer Science, Government, Agriculture. Limitations: Only quantitative data, deals in aggregates, results true on average, misuse possible.

Q12. What do you mean by statistics? Find mean, median, and SD of 100 students:

Marks	0-20	20-40	40-60	60-80	80-100
No. of Students	12	16	35	24	13

Answer:

Statistics: Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting quantitative data to aid decision-making.

Formulas: $\bar{X} = A + (\Sigma fd / \Sigma f) \times h$, $\sigma^2 = [\Sigma fd^2 / \Sigma f - (\Sigma fd / \Sigma f)^2] \times h^2$, $\text{Median} = L + [(N/2 - cf) / f] \times h$

Marks	f	Mid (m)	d	fd	fd ²	CF
0-20	12	10	-2	-24	48	12
20-40	16	30	-1	-16	16	28
40-60	35	50	0	0	0	63
60-80	24	70	1	24	24	87
80-100	13	90	2	26	52	100
Total	100			10	140	

Mean = $50 + (10/100) \times 20 = 52.00$

SD = $\sqrt{[140/100 - (10/100)^2] \times 20} = \sqrt{556.00} = 23.58$

$N/2 = 50$; cf reaches 50 at class 40-60 (cf before=28, f=35)

Median = $40 + (50-28)/35 \times 20 = 52.57$

Q13. Define correlation. Karl Pearson's r for Maths vs Basic Statistics (10 students):

Maths	55	70	40	30	90	80	60	80	90	80
Basic Statistics	65	40	30	50	60	70	50	50	60	70

Answer:

Correlation: A statistical measure expressing the extent to which two variables are linearly related.

Formula: $r = [n \cdot \Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{\{[n \cdot \Sigma x^2 - (\Sigma x)^2][n \cdot \Sigma y^2 - (\Sigma y)^2]\}}$

x	y	x ²	y ²	xy
55	65	3025	4225	3575
70	40	4900	1600	2800
40	30	1600	900	1200
30	50	900	2500	1500
90	60	8100	3600	5400

80	70	6400	4900	5600
60	50	3600	2500	3000
80	50	6400	2500	4000
90	60	8100	3600	5400
80	70	6400	4900	5600
675	545	49425	31225	38075

$$r = 12875 / \sqrt{(38625 \times 15225)} = 12875 / 24250.06 = 0.5309$$

$\therefore r \approx 0.531$ — Positive moderate correlation.

Q14. Define regression. Age vs maintenance cost of computers. Regression equation + estimate for 10 yrs:

Age (years)	2	4	6	8	10
Maintenance cost (000Rs)	10	15	22	32	46

Answer: [Same data as 2021 Q14]

Regression: A statistical method to model the relationship between dependent and independent variables.

From calculations (same as 2021 Q14):

$$b = 4.4500, a = -1.7000$$

$$\text{Regression Line: } Y = -1.7000 + 4.4500X$$

$$\text{When } X=10: Y = -1.7000 + 4.4500 \times 10 = 42.80 \text{ (000 Rs)}$$

Slope $b=4.4500$: For each additional year of age, maintenance cost increases by 4.4500 (000 Rs).

Q15. Define Poisson distribution. Defective lens probability 1/500. Packet of 10. Find $P(0 \text{ defective})$, $P(\geq 1)$, $P(\leq 2)$:

Answer:

Poisson Distribution: A discrete probability distribution expressing the probability of a given number of events in a fixed interval, when events occur with a known constant mean rate.

$$\text{Formula: } P(X=r) = e^{(-\lambda)} \cdot \lambda^r / r!$$

$$n=10, p=1/500=0.002 \rightarrow \lambda=n \times p=10 \times 0.002=0.02$$

$$e^{(-0.02)} = 0.980199$$

$$i) P(X=0) = e^{(-0.02)} \times 1/1 = 0.980199 \approx 0.9802$$

$$ii) P(X \geq 1) = 1 - P(X=0) = 1 - 0.980199 = 0.019801$$

$$iii) P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= 0.980199 + 0.019604 + 0.000196 = 0.999999$$

Q16. Dean wants 95% confidence, error ≤ 0.25 min, $\sigma=1.40$ min. Find required sample size:

Answer:

$$\text{Formula: } n = (Z \cdot \sigma / E)^2$$

$$Z=1.96 \text{ (95\% confidence), } \sigma=1.40, E=0.25$$

$$n = (1.96 \times 1.40 / 0.25)^2 = (10.9760)^2 = 120.47$$

$\therefore$ Required sample size = 121

Q17. Define sampling. Population = {2,8,14,20}. All samples of size 2 without replacement:

Answer:

Population: {2,8,14,20}, N=4, n=2, Samples=C(4,2)=6

Population mean = (2+8+14+20)/4 = 11.0

Sample	Values	Sample Mean
(2, 8)	(2, 8)	5.0
(2, 14)	(2, 14)	8.0
(2, 20)	(2, 20)	11.0
(8, 14)	(8, 14)	11.0
(8, 20)	(8, 20)	14.0
(14, 20)	(14, 20)	17.0
Mean of sample means		11.0000

Mean of all sample means = 11.0000 = Population mean (11.0) ✓

∴ Sample mean is an unbiased estimator of population mean.

Group C [2×10 = 20] — Attempt any TWO questions

Q18. Two computer manufacturers A and B — mean, SD, which is more consistent:

Time (Seconds)	0-2	2-4	4-6	6-8	8-10	10-12
No of A computers	16	13	7	5	4	—
No of B computers	7	12	19	9	1	2

Answer:

Formula: $X_{\square} = \Sigma fm / \Sigma f$, $\sigma = \sqrt{\Sigma f(m - X_{\square})^2 / \Sigma f}$, $CV = (\sigma / X_{\square}) \times 100$

Firm A:

Class	f_A	mid	fm	f(m-X _■) ²
0-2	16	1	16	106.3190
2-4	13	3	39	4.3398
4-6	7	5	35	14.1590
6-8	5	7	35	58.5580
8-10	4	9	36	117.6020
Total	45		161	300.9778

$X_{\square_A} = 3.5778$, $\sigma_A = 2.5862$, $CV_A = 72.2849\%$

Firm B:

Class	f_B	mid	fm	f(m-X _■) ²
0-2	7	1	7	92.7472
2-4	12	3	36	32.2752
4-6	19	5	95	2.4624
6-8	9	7	63	50.1264
8-10	1	9	9	19.0096
10-12	2	11	22	80.8992

Total	50		232	277.5200
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$$\bar{X}_B = 4.6400, \sigma_B = 2.3559, CV_B = 50.7743\%$$

$CV_B (50.77\%) < CV_A (72.28\%) \rightarrow$ **Company B's computers are more consistent.**

Q19. Define normal distribution. Electronic component: mean=5000 hrs, SD=1000 hrs. Find probabilities:

Answer:

Normal Distribution: A continuous probability distribution that is symmetric about the mean, forming a bell-shaped curve. Defined by μ (mean) and σ (standard deviation).

Standard Normal Distribution: A normal distribution with $\mu=0$, $\sigma=1$. Any normal variable can be standardized using $Z=(X-\mu)/\sigma$.

$$\mu=5000, \sigma=1000, Z=(X-5000)/1000$$

Part	Z-calculation	Working	Answer
a) $P(X < 5012)$	$Z = (5012 - 5000)/1000 = 0.012$	$\Phi(0.012) \approx 0.5048$	0.5048
b) $P(4000 < X < 6000)$	$Z_1 = -1.00, Z_2 = 1.00$	$\Phi(1) - \Phi(-1) = 0.8413 - 0.1587$	0.6826
c) $P(X > 7000)$	$Z = (7000 - 5000)/1000 = 2.00$	$1 - \Phi(2.00) = 1 - 0.9772$	0.0228

Q20. Properties of design of experiment. One-way ANOVA — 3 brands of batteries:

Brand A	40	30	50	50	30	
Brand B	60	40	55	65	—	
Brand C	60	50	70	65	75	40

Answer:

Properties of Design of Experiment:

1. **Replication:** Repeating each treatment more than once to estimate experimental error.
2. **Randomization:** Assigning treatments randomly to experimental units to avoid bias.
3. **Local Control:** Grouping experimental units into homogeneous blocks to reduce variability.

One-Way ANOVA:

$$T=780, N=15, CF=40560.00$$

$$TA=200(n=5), TB=220(n=4), TC=360(n=6)$$

$$SSB=8000.00+12100.00+21600.00-40560.00=1140.00$$

Source	SS	df	MS	F-cal	F-tab(2,12) at 5%
Between	1140.00	2	570.00	4.27	3.89
Within	1600.00	12	133.33		
Total	2740.00	14			

$F\text{-cal} (4.27) > F\text{-tab} (3.89) \rightarrow$ **Reject $H_0 \rightarrow$ Brands have significantly different average lifetimes.**

QUICK FORMULA REFERENCE

Formula/Concept	Expression
■ MEASURES OF CENTRAL TENDENCY	
Arithmetic Mean (Grouped)	$\bar{X} = A + (\Sigma fd / \Sigma f) \times h$
Median	$Md = L + [(N/2 - cf) / f] \times h$
Mode	$Mo = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$
Empirical Relation	Mode = 3 · Median – 2 · Mean
■ DISPERSION	
Standard Deviation	$\sigma = \sqrt{[\Sigma f(x - \bar{X})^2 / \Sigma f]} \text{ or } \sqrt{[(\Sigma fd^2 / \Sigma f) - (\Sigma fd / \Sigma f)^2] \times h}$
Coefficient of Variation	$CV = (\sigma / \bar{X}) \times 100\%$
Combined Mean	$\bar{X}_{12} = (n_1 \cdot \bar{X}_1 + n_2 \cdot \bar{X}_2) / (n_1 + n_2)$
Combined Variance	$\sigma_{12}^2 = [n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)] / (n_1 + n_2)$
■ PARTITION VALUES	
Quartile Q1	$L + [(N/4 - cf) / f] \times h$
Decile D7	$L + [(7N/10 - cf) / f] \times h$
Percentile P80	$L + [(80N/100 - cf) / f] \times h$
■ CORRELATION & REGRESSION	
Karl Pearson's r	$r = [n \cdot \Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{[n \cdot \Sigma x^2 - (\Sigma x)^2][n \cdot \Sigma y^2 - (\Sigma y)^2]}$
Regression b _{yx}	$b = (n \cdot \Sigma xy - \Sigma x \cdot \Sigma y) / (n \cdot \Sigma x^2 - (\Sigma x)^2)$
Regression a	$a = \bar{y} - b \cdot \bar{x}$
Regression Line	$Y = a + bX$
Coeff of Determination	r^2 (% variation explained)
■ PROBABILITY DISTRIBUTIONS	
Binomial P(X=r)	$C(n,r) \cdot p^r \cdot q^{(n-r)}, q=1-p$
Poisson P(X=r)	$e^{(-\lambda)} \cdot \lambda^r / r!, \lambda = n \cdot p$
Normal Z	$Z = (X - \mu) / \sigma$
■ SAMPLING	
Sample size (mean)	$n = (Z \cdot \sigma / E)^2$
Sample size (proportion)	$n = Z^2 \cdot p \cdot q / E^2$
■ ANOVA (One-Way)	
CF	T^2 / N
SSB (Between)	$\Sigma (T_i^2 / n_i) - CF$
SST (Total)	$\Sigma \Sigma x_{ij}^2 - CF$
SSW (Within)	$SST - SSB$
MSB	$SSB / (k - 1)$
MSW	$SSW / (N - k)$
F	MSB / MSW

Standard Normal Table Key Values:

Z	0.00	0.25	0.50	0.75	1.00	1.25	1.28	1.50	1.645	1.96	2.00	2.33	2.576	3.00
$\Phi(Z)$	0.5000	0.5987	0.6915	0.7734	0.8413	0.8944	0.9000	0.9332	0.9500	0.9750	0.9772	0.9901	0.9950	0.9987

F-Table Critical Values (5% significance):

df1 \ df2	6	9	12
2	5.14	4.26	3.89
3	4.76	3.86	3.49